When constant g is not enough

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Recognize the limits of the near-surface mgh model.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 5.*
22 J, using $K_f=K_i+W_{net}$.

02. Worked example: explain the model
Field changes with distance; surface-g approximation is poor over planet-scale heights.

03. Read and interpret the model
Book lift; 1 m is tiny relative to radius. Satellite field changes strongly.

04. Apply the physics
Overestimate; actual gravitational pull decreases along path.

05. Explain your reasoning
r=R+h; using h ignores radius and gives wrong field/energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 10.*
6 m/s west and 8 m/s south.

07. Exit check
Constant g; height change tiny relative to Earth radius.

08. STRETCH: test the model
Small altitude is closer to constant 12; at h=R, field 3 N/kg.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 5.*
Net work 37.5 J; v_f=5 m/s from ½mv^2=W_{net}.

02. Worked example: explain the model
Field changes with distance; surface-g approximation is poor over planet-scale heights.

03. Read and interpret the model
Book lift; 1 m is tiny relative to radius. Satellite field changes strongly.

04. Apply the physics
Exact 10 J; constant g estimate 20 J; overestimates.

05. Explain your reasoning
r=R+h; using h ignores radius and gives wrong field/energy.

06. Review: connect an earlier idea *Review: Unit 3, Day 10.*
Other p=(20, -6) kg·m/s; magnitude≈20.88 kg·m/s.

07. Exit check
1 and 0.25 N/kg; strong variation invalidates constant g over this interval.

08. STRETCH: test the model
Exact 200(1/5-1/10)=20 J. Surface g=4; mgh=2×4×5=40 J; approximation overestimates.